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Automated Scoring of Creative Problem Solving With Large Language Models: A Comparison of Originality and Quality Ratings

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Creative problem solving is a naturalistic form of creative thinking involving the generation of solutions that are not only original but also of high quality (i.e., plausible and effective). Past work has shown that large language models (LLMs) can predict human originality ratings of responses to creativity tests. We extend this work to creative problem solving, examining whether both originality and quality can be automatically scored for a naturalistic creativity task. We gathered data from 10 studies, amounting to 3,243 participants who completed different items of the creative problem-solving task (CPST). We then fine-tuned two open-source LLMs, RoBERTa and GPT-2, and few-shot prompted two separate LLMs, Claude and Llama, to predict human ratings of originality and quality on the CPST. We compared LLM performance to two other scoring methods: elaboration and semantic distance. We found that RoBERTa and GPT-2 models predict human ratings of solution quality (RoBERTa, $r = .83$; GPT-2, $r = .83$) better than solution originality (RoBERTa, $r = .79$; GPT-2, $r = .80$). Moreover, we found that both models outperformed elaboration and semantic distance and generalized to new CPST items not in their training set, with stronger predictions for quality than originality on the holdout-prompt set. Few-shot prompting was less effective than fine-tuning at predicting both originality ($r = .66$–$.11$) and quality ($r = .62$–$.26$). We show for the first time that naturalistic creativity tasks can be automatically scored for both originality and quality. Open access is provided to the models and training data.

Keywords: automated scoring, creativity, creative problem-solving, large language models, naturalistic creativity assessment

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Creativity measurement has undergone a rapid evolution, originating in frequency-based methods (Torrance, 1972) and subjective human ratings (Amabile, 1982; Silvia et al., 2008), and more recently, automated scoring techniques (Beaty et al., 2021; Beaty & Johnson, 2021; Dumas & Dunbar, 2014; Dumas et al., 2020). Automated creativity scoring leverages advances in natural language processing, avoiding the limitations of human scoring, such as intensive labor costs or disagreement between raters (Barbot, 2018; Forthmann et al., 2017; Reiter-Palmon et al., 2019). So far, these methods have mostly been applied to scoring divergent thinking tasks, such as the alternate uses task (AUT) (Guilford, 1967), and they have focused exclusively on scoring the originality of ideas (Acar et al., 2023; Beaty et al., 2022; Beaty & Johnson, 2021; Heinen & Johnson, 2018; Johnson et al., 2022). Real-world creative problem solving, however, often requires both originality and quality (i.e., plausible and effective) to ensure effective implementation of creative solutions (Reiter-Palmon, 2020). Yet no methods currently exist for automatic scoring of such naturalistic creativity tasks. In the present study, we extend research on automatic creativity scoring to real-world creative problem solving (Reiter-Palmon et al., 1998, 2009, 2012), aiming to develop more predictive methods for evaluating the originality and quality of creative solutions.

Among the most notable advancements in the scoring of creative products or ideas is the development of the consensual assessment technique (CAT; Amabile, 1982, 1996; Baer et al., 2004). Having undergone decades of refinement, the CAT is considered a gold standard for creativity assessment across many domains (Baer & McKool, 2009). The main principle guiding the CAT is that the combined opinions of several experts from a particular field will

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approximate the true creative value of a product or idea (Kaufman et al., 2009). However, a clear limitation of the CAT is that of high labor costs, exemplified by the difficulty of gathering domain experts to rate numerous responses to a creativity test (Kaufman et al., 2005). Although mixed evidence suggests that novice raters may perform comparably to domain experts (Kaufman et al., 2005, 2013; Silvia et al., 2008), issues such as high labor costs and interrater disagreement are unavoidable with human raters, illustrating the need for a shift toward automated creativity scoring (Paulus et al., 1970).

Automated scoring of creativity dates back to early text-mining approaches developed by Paulus et al. (1970). These first attempts used linguistic variables such as elaboration (i.e., word count) as predictors for the creativity of an idea (Paulus & Renzuli, 1968). While positive correlations have been reported between elaboration measures and creativity ratings, weak prediction strengths have excluded the replacement of human rating methods (Johnson et al., 2022; Taylor & Barbot, 2021). Automated creativity scoring was only later pursued with the introduction of semantic distance, an aspect of originality reflecting remote relationships between concepts. The distributional semantics theory holds that a word's meaning can be determined by examining the context in which it appears (Firth, 1957). Essentially, words that appear in the same context (e.g., sentence) are thought to hold similar meanings, leading to the assumption that word meanings can be extracted from distributions of co-occurrences between words in natural language (e.g., books; Lenci, 2018). Just as the words "dog" and "cat" can be expected to co-occur in text, because of their semantic relatedness, the words "dog" and "plane" will be less likely to co-occur, due to their semantic unrelatedness.

Such comparisons between word meanings can then be mathematically quantified with word embeddings—vector representations of words in a high-dimensional space (Landauer et al., 1998). Numerous methods for generating word embeddings have been developed, with some early approaches involving latent semantic analysis (LSA; Deerwester et al., 1990) and word embedding models such as Word2Vec (Mikolov et al., 2013) or GloVe (Pennington et al., 2014). LSA determines the semantic similarity between pairs of words by computing the probability of them co-occurring in a given context (e.g., a document of arbitrary length). On the other hand, Word2Vec and GloVe involve small neural networks which are trained to predict the meaning of words (e.g., dog) given the context they appear in (e.g., The _ runs).

The application of distributional semantics to creativity assessment is consistent with the associative theory of creativity, which holds that creative ideas arise from the combination of semantically "distant" concepts from memory (Beaty & Kenett, 2023; Kenett, 2019; Mednick, 1962). In this way, it was proposed that the originality of an idea or creative product could be determined by quantifying the semantic distance—derived from the cosine of the angle between pairs of word embeddings in vector space—of the concepts composing it (Beaty & Johnson, 2021). Research has demonstrated that semantic distance predicts human ratings of originality on the AUT, a widely adopted creative thinking task that requires participants to produce unusual uses for everyday objects (Beaty et al., 2022; Dickson et al., 2022; Dumas & Runco, 2018; Forthmann et al., 2022; Maio et al., 2020; Patterson et al., 2023). For the AUT, semantic distance is computed between the everyday object posed to participants (e.g., box) and all words in the participant's response (e.g., seat cushion). At the participant level, strong correlations have been reported between semantic distance scores and human originality ratings on the AUT ($r = .84$; Beaty & Johnson, 2021), although performance has been noted to drop when correlations are computed at the response level ($r = .20$; Organisciak et al., 2022).

Recently, divergent semantic integration (DSI) was developed to compute semantic distance in longer text responses, such as short stories (Johnson et al., 2022). DSI computes the semantic distance between all possible pairs of words using embeddings from pretrained large language models (LLMs), a class of deep neural network models trained on vast textual data for understanding and generating language. Unlike prior vector space models, LLMs are sensitive to the context in which words appear and thus words with multiple meanings—also known as polysemy (e.g., computer server, restaurant server)—making them more suitable for evaluating long-form creative responses than previous semantic distance approaches that do not consider context (e.g., LSA; Jawahar et al., 2019). Indeed, Johnson et al. (2022) found that word embeddings extracted from LLMs allowed for the best predictions of human-scored creativity ratings of short stories, explaining up to 72% of the variance in human ratings. Of note, the model also demonstrated good predictions when faced with responses from nonnative English speakers, compared to those from native English speakers, providing support for its generalizability.

Importantly, semantic distance metrics are "unsupervised" text-mining approaches, calculating similarity metrics without human involvement in the process. Other models, leveraging supervised learning, are being increasingly adopted for their superior performance across a range of linguistic tasks, such as sentence completion, sentiment analysis, and question answering (Devlin et al., 2018; Y. Liu et al., 2019; Radford et al., 2018; Raffel et al., 2020; Vaswani et al., 2017; Yang et al., 2019). Supervised learning involves training a machine learning model on a labeled data set (i.e., the "training set"), such as responses to a task and originality scores, so that the model can learn to predict new labels it has not seen before. In a first attempt at applying supervised learning to automate creativity scoring, researchers trained several machine learning classifiers on a variety of text features (e.g., word embeddings, word length, word count) and compared their performance at predicting human creativity ratings on the AUT (Buczak et al., 2022). Researchers observed how their best model would explain between 33% and 56% of the variance in the AUT scores. This work built directly on past research employing word embeddings/semantic distance by expanding on the types of models and data that were employed in the prediction of creativity. Further work has also successfully applied semantic distance measures for scoring the flexibility of generated ideas, operationalized in terms of the amount and magnitude of switches between semantic categories across a number of ideas (Grajzel et al., 2023; Gray et al., 2019).

LLMs have recently shown performance far superior to semantic distance metrics on the AUT (Organisciak et al., 2022; Stevenson et al., 2022). LLMs have garnered attention more broadly for achieving human-level performance across many complex tasks, from language translation and text generation to sentiment analysis and question answering (Ali et al., 2022; Hendy et al., 2023; Kheiri & Karimi, 2023). LLMs typically undergo significant pretraining on general language data (i.e., large volumes of text), and are later "fine-tuned" on smaller, specific data sets (e.g., creative responses and

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human ratings) to boost their performance on related tasks (e.g., creativity scoring; P. Liu et al., 2023). Fine-tuning is a supervised approach that allows LLMs to build on their vast knowledge obtained during pretraining by adjusting their parameters through training on additional domain-relevant examples (e.g., AUT responses and human ratings). LLMs such as the generative pretrained transformer (GPT) are large and complex, with millions to billions of parameters, which also tend to increase exponentially with every new version of the model (e.g., GPT-2, GPT-3, GPT-4).

Thus, despite their impressive capabilities across a variety of benchmarks (Y. Wang et al., 2024), LLMs have poor interpretability (i.e., they are a "black box"; Barredo Arrieta et al., 2020; Dale, 2021; Gunning et al., 2019; Kojima et al., 2022), unlike semantic distance, which has a clearer interpretation. While it may be possible to interpret the semantic distance score between two words as their dissimilarity within a conceptual space, it would be complicated to draw similar conclusions about how LLMs draw predictions regarding the creativity of an idea. As such, fine-tuned LLMs may currently provide only limited insights as to how humans evaluate the creativity of ideas. Nevertheless, LLMs can still serve as a good solution for the practical challenge of automating creativity scoring, where computational methods are benchmarked according to their performance at predicting human ratings.

Indeed, compared to semantic distance, the performance gains of LLMs are substantial. When trained on 27,000 AUT responses, Organisciak et al. (2022) found that GPT-3 robustly predicted human creativity ratings of responses it had not seen before ($r = .81$) compared to semantic distance ($r = .20$). Organisciak and colleagues offered practical insights into how fine-tuning could be applied to other creativity tasks, and the requirements for reliable prediction of human ratings. For example, as the size of the training data set was reduced, the performance of their models decreased. Yet strong performance ($r = .75$) was found to require about 6,000 responses; this suggests that successful fine-tuning can be achieved without very large data sets that are difficult to obtain and manually score. Organisciak et al. also demonstrated that few-shot learning—when a model is prompted to complete a task with only a small number of training examples—can be applied to AUT scoring, with GPT-4 demonstrating good predictions of human ratings ($r = .70$) when provided only 20 example responses.

While no attempts have been made to automatically score the quality of ideas in the context of the CAT or creativity tasks, some related work has investigated the performance of LLMs for the automatic scoring of the quality, or "correctness," of creative products such as essays (Pack et al., 2024) or peer reviews (Zhou et al., 2024). Considerable attention has been given to modern LLMs and their potential for transforming educational settings, particularly in terms of automating largely algorithmic tasks such as essay marking (Bahroun et al., 2023; Ke & Ng, 2019). In one study, fine-tuned LLMs have been shown to moderately predict human ratings of quality in a large data set of 27,909 student essays ($r = .19$–$.46$; Morris et al., 2023). Of note, such essays are markedly longer, and more linguistically complex, than the typical responses to creativity tasks such as the AUT. Further, essay marking relies on different rubrics and scoring techniques than those employed in the CAT, which is commonly employed for scoring responses to creative tasks. As such, it is unclear whether such findings will extend to the context of automatic creativity scoring.

The Present Research

Automatic creativity scoring has largely focused on scoring the originality of ideas, with most work involving divergent thinking tasks such as the AUT (Beaty & Johnson, 2021; Johnson et al., 2022; Organisciak et al., 2022; Stevenson et al., 2022). To our knowledge, no such work has investigated approaches to automatically scoring the quality of ideas. Importantly, originality and quality are distinct components of creativity with unique relationships to other constructs (Morral-Robinson & Reiter-Palmon, 2013; Reiter-Palmon et al., 1997, 2009). Further, divergent thinking tasks, such as the AUT, often have little resemblance to problems encountered in real life. Creative problem-solving tasks, in contrast, involve the production of a solution that is then evaluated for its originality and quality. These tasks can range from navigating social dilemmas (e.g., resolving a social dispute) to addressing job-related matters (e.g., crafting a marketing strategy). Given the broad importance of real-world creative problem solving (Reiter-Palmon et al., 1998, 2009, 2012), it is important to develop efficient ways to score such naturalistic tasks, which are limited by the same bottlenecks of scoring divergent thinking tasks (e.g., high human labor cost).

To address this issue, we examined data from the creative problem-solving task (CPST)—a paradigm involving a naturalistic, real-world expression of creative thinking—and tested whether it could be scored automatically by using LLMs. We compiled data from 10 studies, providing us with a total of 3,243 participants who completed different problems of the CPST. Responses were rated by human judges for both originality and quality, allowing us to train LLMs to predict each metric separately. We thus investigated, for the first time, whether the quality of creative responses can be predicted with automated scoring methods. We fine-tuned two open-source LLMs, RoBERTa and GPT-2, on originality and quality human ratings to responses on the CPST. We also ran few-shot prompting with two state-of-the-art LLMs, Claude-3-Haiku and Llama-3.1-70b, and compared this to the fine-tuning to test different automated creativity scoring techniques. To evaluate model performance, predictions were compared to two other scoring methods—elaboration (i.e., word count) and semantic distance—which reliably predict human creativity ratings in divergent thinking tasks but have not yet been evaluated in the context of creative problem solving.

Method

The data sets, analysis code, and few-shot prompts are available as the additional online materials on the Open Science Framework (Luchini et al., 2024).

Participants

The data for this project were compiled from a set of 10 previous studies conducted at U.S. universities. Of these, nine recruited only students enrolled in an introductory psychology course at large midwestern universities, and one (Ralph1) was conducted through Amazon MTurk. The total data set spans some published as well as unpublished data sets. The compiled data set included a total sample of 3,243 participants, with each participant providing a single response to a problem from the CPST (see the Procedure section). The total amount of responses was thus equal to the number of participants. Each individual data set included a single version of the

CPST presented to all participants. The task instructions were generally the same, but the problem presented to participants varied. Across all 10 data sets, there are a total of six different problems. The data from the Clara1 study were previously included in an article published in a peer-reviewed journal (Reiter-Palmon et al., 2012).

Procedure

Participants completed several tasks and questionnaires within each study from which the data were extracted. Each study involved its own unique combination of tasks and order of presentation; only the CPST is considered here.

CPST

Participants were presented with a problem scenario and tasked to generate a single solution to solve it. Each problem scenario: (a) described a central character and the everyday problem they were facing (e.g., at school, at work, or in their relationships) and (b) provided sufficient context that highlighted several parameters of the problem (e.g., the characters' motivations, background, relationship with other entities in the scenario; Table 1). Problem scenarios were intentionally designed to be familiar to participants, allowing them to leverage their own expertise and experiences when generating a solution. Additionally, problem scenarios were detailed enough so participants could attend to any specific aspect(s) of the problem, allowing multiple acceptable approaches and no clear or correct solution. Taken together, the structure of the CPST was conducive to generating an original yet high-quality (i.e., creative) solution (Reiter-Palmon et al., 1998, 2009).

CAT

Responses were scored by human raters on two metrics: originality and quality. The rating procedure was adapted from the CAT (Amabile, 1982) developed by Reiter-Palmon (2020). For each data set, all solutions were rated by a set of two to four raters for originality, while a different set of raters rated the same solutions for quality. Raters underwent a group training where they were instructed on how to evaluate responses for either originality or quality. During the training, emphasis was placed on defining either originality or quality to ensure raters had a shared understanding of their assigned metric's meaning. Raters then took part in a practice round of scoring responses, where each rater individually scored a set of responses before sharing them with the group. Discrepancies between raters were then discussed to ensure a common understanding of the rating scale. A highly original solution was defined as one that is unique and novel, imaginative, and not structured by the problem. A high-quality solution was defined as one that addressed two or more issues posed by the problem and effectively addressed them. Scores were provided on a 5-point Likert-type scale (5 = *very original/very high quality*; 1 = *very unoriginal/very low quality*).

Following the training, raters were asked to independently rate 10% of solutions to the problem they were trained on. Interrater reliability was evaluated on this 10% set to ensure consistency in ratings. If raters demonstrated poor interrater reliability (i.e., intraclass correlation coefficient (ICC) $< .60$; Shrout & Fleiss, 1979), raters would undergo additional training sessions until a satisfactory level of interrater reliability was achieved. However, across studies, all raters achieved satisfactory interrater reliability, so no additional training was necessary. Finally, raters were asked to rate 100% of solutions to the problem they were trained on. For interrater reliability results on 100% of responses across each data set, see Table 4.

For the present study, factor analysis was used to obtain a single composite score for automated scoring. Factor analytic methods, such as confirmatory factor analysis (CFA), have been widely applied to improve reliability and validity of subjective creativity scoring methods by identifying common variance across raters (Silvia, 2011). Separately for originality and quality, a single factor score was calculated from each response, by extracting a weighted composite of all individual rater scores using CFA (Table 2). The CFA was constructed as a unidimensional model with each individual rater as an observed variable. The analysis was run in R (Version 4.2.1), using R studio (Version 2023.03.0) and by employing the lavaan package (Version 0.6).

Table 1
Wording of CPST Problems

Problem ID	Problem
Becky	Becky is a college student who works part-time at Mark's Pizzeria. Mark, the owner of the restaurant, has treated Becky very well. He gave her a job that she needs to help pay her rent when no other business would employ her because she was arrested for shoplifting 3 years ago. Mark also lets Becky work around her school schedule, and has asked if she wants to be a shift manager in the summers. Becky's roommate Jim also works at the pizzeria, but Jim has been causing a lot of problems at work. He always avoids doing his job, treats customers rudely, and makes a lot of mistakes with orders. Jim recently began stealing food from the pizzeria. Two days ago the pizzeria was short-staffed, so Jim and Becky were the only employees left at closing time. Jim made 10 extra pizzas and took them home to a party he was hosting without paying for them. Becky feels like she needs to do something about Jim's behavior. However, Becky is hesitant to tell Mark about Jim because Jim is a good friend to Becky. Becky also needs Jim to have a job so he can pay his portion of their rent. Becky does not know what to do.
Mike	Mike is a graduate student at a Midwestern university. He has a problem with his transportation to school because he just moved to an apartment on a steep hill. Unfortunately for Mike, when it snows his sports car cannot get enough traction to make it up the hill to his garage. Currently, he gets a ride from his wife when he cannot drive, but she is tired of driving him around. Mike does not want to sell his car because it was his dream car since he was a little kid, and now that he has had it for 6 years; it is worth more to him than he can sell it for. Also, Mike has a very small income as a graduate student, and getting another job to pay for a new vehicle could jeopardize his education. His wife's uncle is a car salesman who has found a couple sports utility vehicles that would solve his problem, but another local car dealer has the same vehicles for a cheaper price. Mike's wife wants to buy a sports utility vehicle from her uncle because he will assure the quality of the vehicle.

Note. CPST = creative problem-solving task.

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Table 2
Statistics for Confirmatory Factor Analysis

Data set	χ^2 originality (p)	CFI originality	RMSEA originality	χ^2 quality (p)	CFI quality	RMSEA quality
Acme1	.23	1	.05	.18	.99	.06
Acme2	.44	1	0	.07	.99	.11
Acme3	.004	.98	.13	<.001	.93	.32
Becky1	<.001	.97	.19	NA	1	0
Becky2	.003	.97	.21	.03	.99	.14
Becky3	.05	1	.08	<.001	.7	.47
Clara1	<.001	.99	.14	<.001	.97	.26
Joan1	<.001	.98	.21	.007	.99	.12
Mike1	<.001	.76	.47	.04	.99	.14
Ralph1	<.001	.97	.23	<.001	.89	.38

Note. Each individual data set is named in terms of the specific problem used (i.e., acme, becky, clara, joan, mike, and ralph) and arbitrarily numbered to distinguish different data sets which include the same problem. CFI = comparative fit index; RMSEA = root-mean-square error of approximation; NA = not available.

Automated Scoring of Quality and Originality

DSI

To provide a semantic distance baseline to compare to fine-tuning, DSI was selected, given its ability to predict human creativity ratings on text responses of similar length to the CPST (i.e., short stories; Johnson et al., 2022). DSI uses word embeddings extracted from pretrained models to compute pairwise semantic distance between all words in a text. Because DSI is an unsupervised method (i.e., it does not involve adjusting the pretrained model weights or parameters for a specific task like creativity scoring), it serves as an ideal benchmark for evaluating the performance of supervised/fine-tuned models. Like other semantic distance measures (Beaty & Johnson, 2021), DSI computes semantic dissimilarity via the converse cosine similarity formula (1—semantic similarity). To obtain a DSI score, for each text response to the CPST, the semantic distance values between all possible word pairs in the response are calculated and then averaged. We used word embeddings extracted from the bidirectional encoder representations from transformers (BERT) model, given its context sensitivity and reliable performance in predicting human creativity ratings on other tasks (i.e., short stories; Johnson et al., 2022). DSI was computed using the pipeline developed by Johnson et al. (2022).

Robustly Optimized BERT Pretraining Approach (RoBERTa)

The RoBERTa model is an improvement on the BERT model (Y. Liu et al., 2019), which was developed by Google and open-sourced in 2018 (Devlin et al., 2018). RoBERTa is a transformer model (see Vaswani et al., 2017) pretrained via a self-supervised approach, without any human labeling of the natural language data. RoBERTa maintains a similar architecture to BERT but implements several changes in its pretraining procedure, such as an increased amount of data it was trained on (Y. Liu et al., 2019). The base version of the model contains 123 million parameters and was pretrained on a collection of data from the BookCorpus (800 M words), the English portion of Wikipedia (2,500 M words), the CommonCrawl-News English data set (63 M news articles), the OpenWebText data set of Reddit posts, and the Stories data set which contains a subset of the CommonCrawl repository.

GPT-2

The GPT-2 model was developed by OpenAI and open-sourced in 2019 (Radford et al., 2019). GPT-2 is also a transformer model pretrained via a self-supervised approach. The full version of the model contains 1.5 billion parameters and was pretrained on the WebText data set, which aggregates content from eight million web pages, primarily extracted from Reddit outbound links.

Fine-Tuned Models

Following best practices in machine learning, responses from each data set were randomly separated into training and holdout-responses data following an 80/20 ratio (Raschka & Mirjalili, 2019). In this way, the models were used to predict creativity scores for the holdout-responses (i.e., test set) only after the training procedure was completed. This approach is common in machine learning research, which ensures that the generalizability of the models could be tested on a batch of unseen responses. Further, the Mike problem (i.e., the entire mike1 data set) was withheld from this splitting procedure and included as a holdout-item set. This method ensured that model generalizability could also be tested toward problems it was never presented during training. If the models generalize to new responses and prompts, they have likely learned something general about how humans evaluate originality and quality on this task, and would allow for new prompts to be scored by the models.

Two open-access LLMs were selected for fine-tuning, RoBERTa and GPT-2, which belong to the BERT and GPT families, respectively. RoBERTa and GPT-2 notably differ in terms of their architecture and training objectives. For instance, RoBERTa was trained with a masked language modeling approach (i.e., the model predicts missing words from a sequence with access to both left and right context) aimed toward understanding context (i.e., filling in the blanks), while GPT-2 underwent autoregressive training (i.e., the model predicts missing words left to right, following a language modeling objective) for generating text (i.e., creating sentences). Employing two models with different architectures and training allowed us to get a broader coverage of the LLM space. The two families of models have further been shown to perform strongly on a variety of linguistic tasks such as question answering and reading comprehension (Gillioz et al., 2020). All fine-tuned models employed in the present analysis are open-accessible on the Huggingface platform (https://huggingface.co/).

For our analysis, we selected the RoBERTa-base model, which contains 123 million parameters, and the small version of the GPT-2 model which contains 137 million parameters. These versions were selected in favor of larger variants of the same models given their comparable sizes and the reduced computational resources needed for fine-tuning models with fewer parameters. We obtained the pretrained models from the "Huggingface Transformers" suite was used via the Python programming language (Wolf et al., 2020). Fine-tuning was performed twice for each model, separately for originality and quality, such that the fine-tuned models would be specialized in the prediction of either score. In total, four fine-tuning procedures were employed for the subsequent prediction of originality and quality scores by both RoBERTa and GPT-2. All fine-tuning procedures used the same hyperparameter settings, selected based on hyperparameter tuning of a related creativity scoring model which was employed for unpublished work. The learning rate was set to .00005, batch size to 16, and a total of 116 training

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epochs were implemented. All other hyperparameters were left to the default settings associated with each model.

Few-Shot Prompting

We employed two state-of-the-art LLMs for our few-shot prompting: Claude-3-Haiku (https://www.anthropic.com/news/claude-3-haiku) and Llama-3.1-70b (Meta, 2024). These two models were both released in 2024, developed by Anthropic and Meta, respectively. We employed these models given that few-shot shot prompting is generally thought to require more advanced LLMs due to the complexity of in-context learning. Few-shot prompting relies on in-context learning, wherein a LLM completes a task given only a few examples and a set of instructions, without the need for additional training or updating of its parameters. In our case, an example consisted of a single CPST item and response, with its associated originality or quality rating.

We varied which item the examples provided to the LLM would belong to, such that during each round of prompting the LLM would only see examples belonging to a single item. This was done to ensure the prompt would fit into the model's context window (the maximum number of words that can be sent to the LLM at once), as CPST items are quite long and we found that including multiple items in the prompt degraded performance. In this way, we were able to determine whether examples from specific items were more effective for maximizing model performance on the holdout-responses or holdout-prompt sets. We pulled all examples from the training set, ensuring that any examples would not be present in either holdout-responses or holdout-prompt sets. Further, we never provided examples associated with the Mike problem, thus ensuring that the holdout-prompt set would serve as a test of generalizability to unseen prompts. We ran our few-shot prompting separately for originality and quality, designing different instructions for each.

Given the high word count for both CPST items and responses, we scored each response individually and instructed the LLMs to only output the predicted originality or quality rating (Table 3). We penalized any model that did not follow the task instruction by assigning a random score for any LLM generation that did not produce an integer value at the end of the output. We repeated our prompting by providing the LLMs with either 0, 5, 10, or 20 examples, to test whether increasing this amount would increase model performance. In terms of hyperparameters, we set the temperature to 0 to reduce stochasticity in the outputs and left all other settings to their default values. For our Llama model, we additionally ran each trial using three random seeds, where each seed selected a different combination of examples to account for any example-specific effects. This was not possible for our Claude model as we observed that employing multiple random seeds during prompting had no effect on the models' output.

For the purposes of few-shot prompting, we transformed the factor scored originality and quality human ratings by splitting responses into quartiles and mapping them to a 4-point scale. Separately for originality and quality, responses were assigned a score of 0 if they fell in the lower quartile (<25%), 1 if in the middle-lower quartile (25%–50%), and 2 for the middle-upper.

Results

We first computed descriptive statistics for the CPST (Table 4). Interrater reliability for originality and quality across all data sets was found to range from satisfactory to excellent (originality ICCs = .71–.96; quality ICCs = .76–.88). In our analysis, we ran a series of Person's correlations between human ratings, of originality and quality, and our baseline measures and fine-tuned LLM predictions. For our few-shot prompting, we instead ran a series of Spearman's correlations between human ratings and model predictions, given the ranked nature of the data. As baseline measures, we selected word count and DSI, to compare LLM performance against the current best methods for automatically scoring the creativity of long-form creativity responses.

We hypothesized that predictions made by fine-tuned LLMs would surpass the baseline measures in their correlations with human ratings. For the pairwise correlations between human ratings and LLM predictions, we also report prediction errors for originality and quality so as to explore specific weaknesses in model performance.

Outliers were identified based on Cook's distance metrics and were thus removed before running each regression model (Cook, 1977). Thresholds for outlier identification were calculated separately for the entire data set (Cook's distance > .0012) holdout-responses set (Cook's distance > .0066) and holdout-prompt set (Cook's distance > .023), based on the modified Cook's cutoff equation (Fox, 2019). Across all data sets, the average word count was 49, and mean ratings were 2.42 for originality and 2.56 for quality. Although we only report analyses on the data with outlier removal, our analyses revealed consistent findings both including and excluding outliers (Appendix).

Automatic Scoring: DSI and Word Count

We ran a series of Pearson's correlations between DSI and word count across all studies included in our final data set (Table 5). We found a positive correlation between originality and quality ratings on the CPST ($r = .51$, $p < .001$, outliers removed = 165), consistent with past creativity research (Arreola & Reiter-Palmon, 2016). We further found a positive correlation between word count and originality ($r = .49$, $p < .001$, outliers removed = 157) and between word count and quality ($r = .68$, $p < .001$, outliers removed = 147), potentially indicating that raters emphasized word count of solutions.

Next, we computed the correlations between DSI and originality and quality for each solution (Table 5). Across the full data set, we found that DSI was moderately correlated with both originality ($r = .39$, $p < .001$, outliers removed = 96) and quality ($r = .44$, $p < .001$, outliers removed = 85). We also found that DSI was correlated with word count ($r = .56$, $p < .001$, outliers removed = 97).

Automatic Scoring: Fine-Tuned LLMs

The fine-tuned LLMs were then used to predict originality and quality scores for each response, separately for each model. We then ran Pearson's correlations between the predicted and human-rated scores, separately for the holdout-responses set (Figure 1) and holdout-prompt set (Figure 2).

For originality, the RoBERTa model robustly predicted human ratings on the holdout-responses set ($r = .79$, $p < .001$, outliers removed = 36). Further, GPT-2 predictions of originality also correlated strongly with the human ratings ($r = .80$, $p < .001$, outliers removed = 35). We then observed that mean prediction errors were typically higher for human-rated originality above or below two standard deviations from the mean than for scores between

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Table 3
Few-Shot Prompt for Originality

Few-shot prompt
You are tasked with rating the originality of solutions to this creative problem-solving task: {problem} These solutions were generated by college students in a psychology experiment. When assessing originality, consider three key aspects: novelty, imagination, and structure. Novelty refers to how unique the approach is compared to typical solutions. Imagination considers how creative and inventive the idea is. Structure examines whether the idea is limited by the scenario as presented, or if it questions assumptions and demonstrates out-of-the-box thinking. Rate the solution as one in its totality, even if there are multiple parts or ideas. Do not "read into" the idea or assume intentions—rate based solely on what is written. Remember that originality is independent from quality—a solution can be highly original but low in quality, or vice versa. Use a 5-point scale for rating, where: 0 is *very unoriginal* (very simple and/or common idea) 1 is *unoriginal* (simple idea, not novel or imaginative, structured by the scenario) 2 is *neutral* (limited novelty or imagination, still structured by the scenario) 3 is *original* (shows some novelty and imagination, less structured by the problem) 4 is *very original* (unique, novel, imaginative, and not structured by the scenario) Here are some example responses and their originality ratings to guide your rating: {examples} For the given solution, provide only a 0–4 originality rating. Do not explain your rating or provide any additional commentary. {response}

Note. For each few-shot round, {problem} was replaced by the relevant item and {response} was replaced with a single CPST response. Further, {examples} was either removed in the case of 0 shot, or replaced with 5, 10, or 20 CPST responses.quartile (50%–75%), and finally a score of 3 for the upper quartile (>75%). Models were thus provided the transformed scores in the examples and were tasked with scoring unseen responses on this 4-point scale. CPST = creative problem-solving task.

two standard deviations from the mean (Table 5); this result suggests the models were overestimating the least original solutions and underestimating the most original solutions (cf. Patterson et al., 2024). For quality, the RoBERTa and GPT-2 quality models showed equally robust predictions of human scores ($r = .83$, $p < .001$, outliers removed = 29–35), which were numerically higher than those for the originality models. As with originality ratings, the mean prediction errors tended to be higher for human quality scores above or below two standard deviations from the mean (Table 6).

In terms of the holdout-prompt set, model performance decreased for both RoBERTa and GPT-2 when predicting originality. The correlation between the RoBERTa model and originality was slightly stronger ($r = .41$, $p < .001$, outliers removed = 11) than that between GPT-2 and originality ($r = .40$, $p < .001$, outliers removed = 9). In terms of quality, RoBERTa and GPT-2 both demonstrated extremely robust correlations with human ratings in the holdout-prompt set ($r = .89$, $p < .001$, outliers removed = 10).

Automatic Scoring: Few-Shot Prompting

We then separately prompted the Claude (Figure 3) and Llama (Figure 4) models to predict human ratings of originality and quality on the holdout-responses and holdout-prompt sets. We repeated this prompting by modifying the item associated with the examples provided to the LLM (Acme/Becky/Clara/Joan/Ralph), as well as modifying the number of examples provided to the LLMs (0/5/10/15/20).

For our Claude model, we observed positive correlations between model predictions and human ratings of both originality and quality, on both holdout-responses and holdout-prompt sets (Figure 3). For originality, the performance of the few-shot Claude model on the

Table 4
Descriptive Statistics CPST Responses by Individual Data Sets

Data set	Number of responses	Mean words per response	Mean originality (*SD*)	ICC originality	Mean quality (*SD*)	ICC quality
Acme1	158	59	3.01 (1.10)	.96	2.53 (0.85)	.86
Acme2	215	74	3.05 (0.91)	.83	3.06 (1.00)	.87
Acme3	450	39	2.07 (0.78)	.71	2.30 (0.81)	.86
Becky1	365	65	2.53 (0.86)	.81	2.38 (0.80)	.87
Becky2	178	43	2.45 (0.91)	.84	2.38 (0.84)	.88
Becky3	475	49	2.29 (1.00)	.87	2.76 (0.62)	.76
Clara1	533	53	2.77 (0.96)	.88	2.49 (0.71)	.88
Joan1	470	44	2.56 (0.88)	.90	2.61 (0.82)	.86
Mike1	179	38	2.60 (0.79)	.76	2.35 (0.84)	.84
Ralph1	220	32	1.89 (0.81)	.78	2.74 (0.75)	.85

Note. CPST = creative problem-solving task; ICC = intraclass correlation coefficient.

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Table 5
Correlations Between Human Ratings of Originality and Quality, Word Count, and DSI

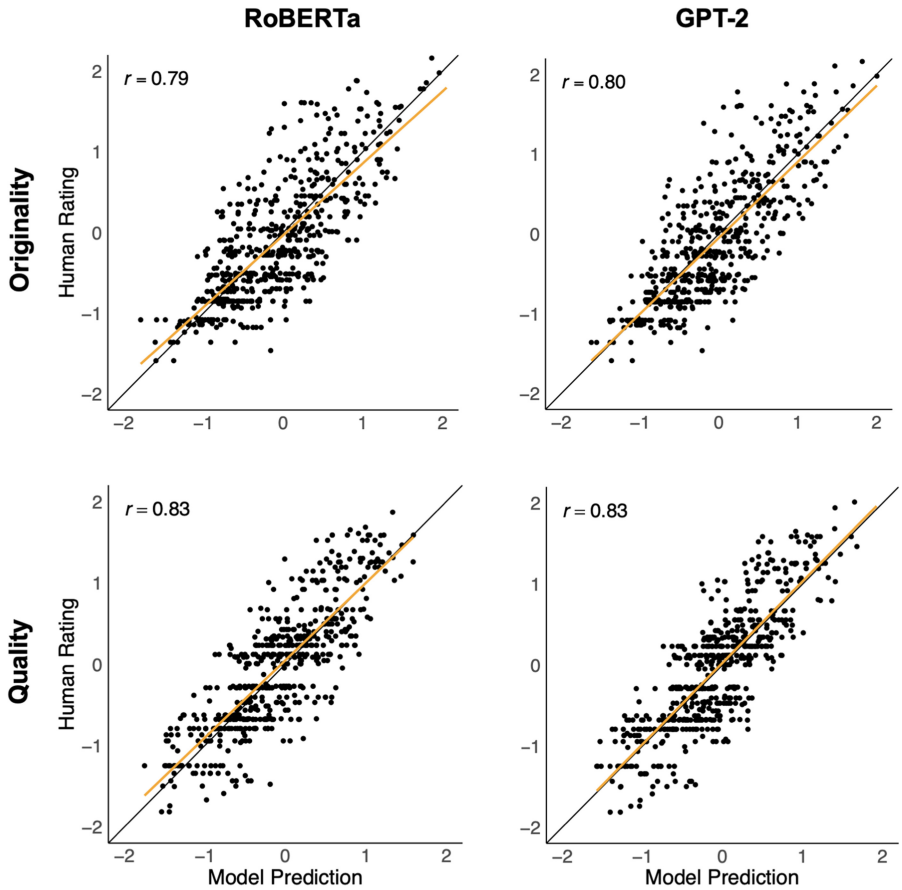
Variable 1	Variable 2	*r*	*p*	Outliers removed
Originality human ratings	Quality human ratings	.51	<.001	165
Originality human ratings	Word count	.49	<.001	157
Originality human ratings	DSI	.39	<.001	96
Quality human ratings	Word count	.68	<.001	147
Quality human ratings	DSI	.44	<.001	85
Word count	Count	.56	<.001	97

Note. DSI = divergent semantic integration.

holdout-responses set was highest when providing the model with 20 examples from the Ralph item ($r = .66$, $p < .001$). Nevertheless, Claude predictions of originality on the holdout-responses set were consistently worse than those achieved by our fine-tuned models. In terms of the holdout-prompt set, we instead observed the highest model performance when providing the model with 10 examples from the Clara item ($r = .43$, $p < .001$), which surpassed the model performance of the fine-tuned models on the same set. For quality we then observed consistently worse model predictions for the few-shot Claude model compared to our fine-tuned models.

We then observed generally positive correlations between our Llama model predictions and human ratings of originality and quality, across the holdout-responses and holdout-prompt sets (Figure 4). Overall, we found the few-shot Llama model performed consistently worse than our fine-tuned models, as well as our few-shot Claude model.

Figure 1
Predicted Human Originality and Quality Scores for RoBERTa and GPT-2 on the Holdout-Responses Set

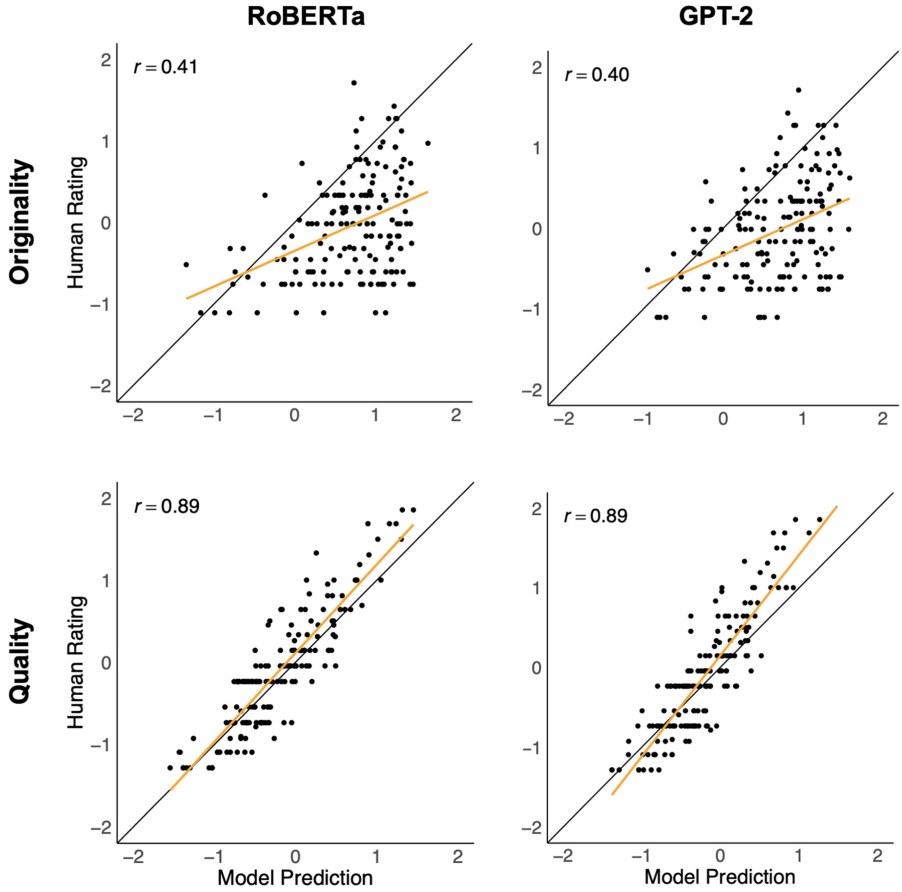


Note. Single responses are denoted by the black dots. Ideal performance ($r = 1.00$) is denoted by the black line, while the orange (dark gray) line is the line of best fit. RoBERTa = robustly optimized bidirectional encoder representations from transformers pretraining approach; GPT = generative pretrained transformer. See the online article for the color version of this figure.

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Figure 2

Predicted Human Originality and Quality Scores for RoBERTa and GPT-2 on the Holdout-Prompt Set



Note. Single responses are denoted by the black dots. Ideal performance ($r = 1.00$) is denoted by the black line, while the orange (dark gray) line is the line of best fit. RoBERTa = robustly optimized bidirectional encoder representations from transformers pretraining approach; GPT = generative pretrained transformer. See the online article for the color version of this figure.

Discussion

Recent advances in machine learning have allowed for the automation of costly human tasks, including manual creativity scoring. Researchers are increasingly using these tools to automatically score verbal creativity tasks, replacing the need for trained human raters and accelerating the pace of creativity research. Nevertheless, automated creativity assessment has focused almost exclusively on scoring the originality of responses to the AUT (Beaty et al., 2021; Beaty & Johnson, 2021; Buczak et al., 2022; Dumas & Dunbar, 2014; Dumas et al., 2020), with no methods currently available to evaluate the quality of responses to tasks that require real-world problem

Table 6

Mean Prediction Errors for Human-Rated Originality and Quality Scores

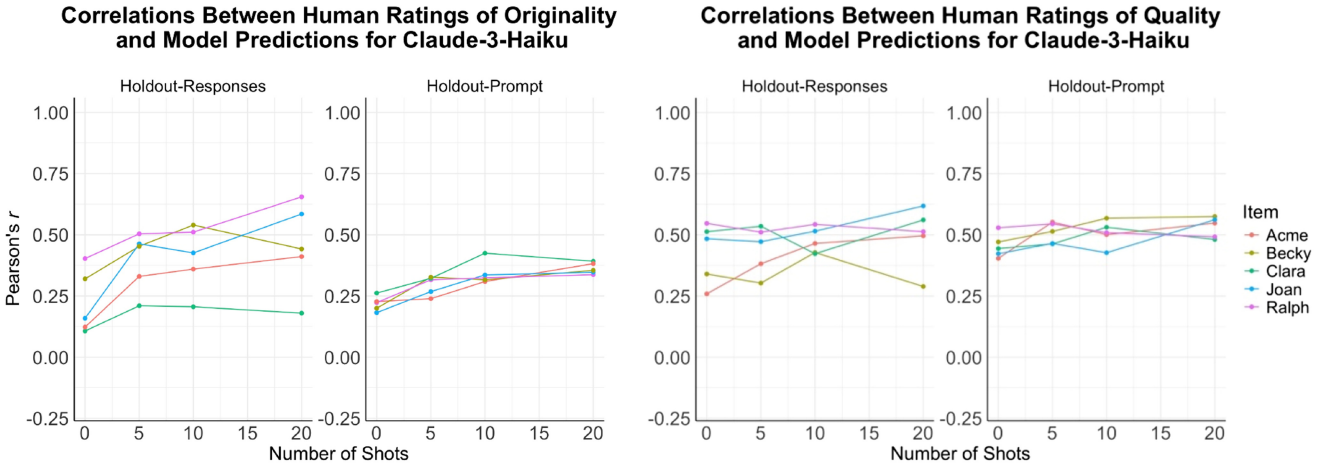
	Originality		Quality	
Data set	±2 *SD*	Within 2 *SD*	±2 *SD*	Within 2 *SD*
RoBERTa holdout-responses set	1.08	0.54	1.02	0.52
RoBERTa holdout-prompt set	1.1	0.81	1.19	0.49
GPT-2 holdout-responses set	1.13	0.51	1.11	0.48
GPT-2 holdout-prompt set	1.06	0.75	0.98	0.42

Note. Errors are calculated over factor scores and are averaged in the form of absolute values, between 2 *SD*s of the mean and above or below 2 *SD*s of the mean. RoBERTa = robustly optimized bidirectional encoder representations from transformers pretraining approach; GPT = generative pretrained transformer.

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Figure 3

Predicted Human Originality and Quality Scores for Few-Shot Claude-3-Haiku on the Holdout-Responses and Holdout-Prompt Sets



Note. Different colored points denote which item the examples were drawn from. Number of shots indicate the amount of examples provided to the model. See the online article for the color version of this figure.

solving. Further, the application of fine-tuned LLMs in automated creativity scoring has focused on short-form responses, with limited naturalistic value. No work to date has attempted to automatically score the creativity (i.e., originality and quality) of responses to the CPST.

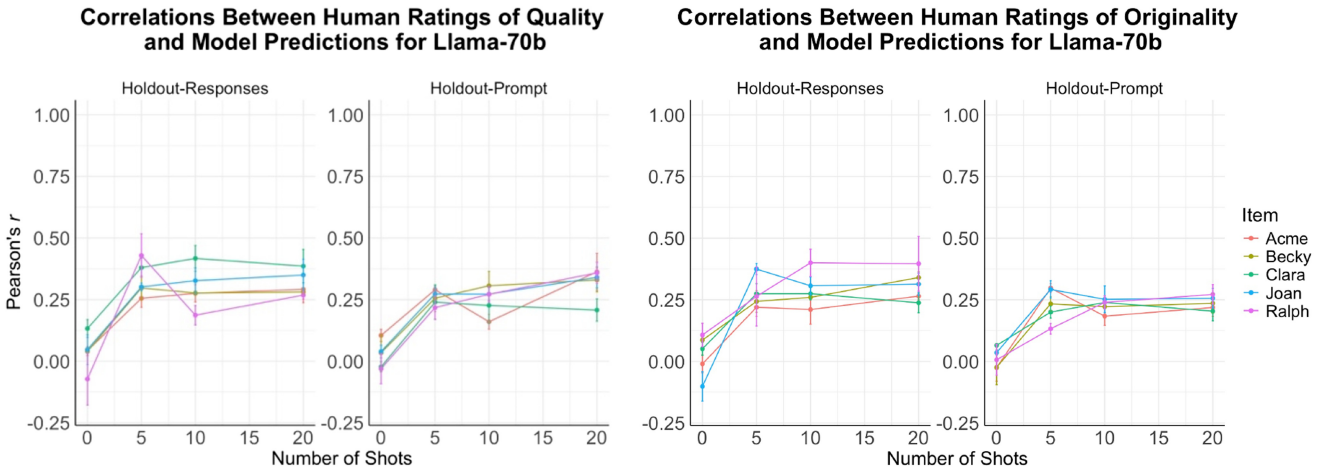
In our analysis, we trained LLMs (i.e., RoBERTa and GPT-2) to predict human-rated originality and quality on the CPST. The models successfully predicted both metrics, yielding substantially higher correlations than those achieved by the best-performing semantic distance model (i.e., DSI). For quality, the performance of the trained LLMs also extended to an "unseen" prompt—problems the models were not exposed to during training—while moderate predictions were observed for originality. Our results demonstrate for the first time that automated scoring methods can capture the quality of responses to real-world creativity tasks, moving beyond the assessment of originality alone.

Correlations between the model predictions and human-rated originality on the trained items were similarly robust for RoBERTa ($r = .79$) and GPT-2 ($r = .80$). This effect size was considerably higher than that of word count ($r = .49$) and double that between originality and DSI/semantic distance scores ($r = .39$). Interestingly, originality predictions dropped considerably for new prompts—a test of model generalization beyond its training set—with GPT-2 ($r = .40$) performing slightly worse than RoBERTa ($r = .41$). Of note, predictions on the originality holdout-prompt set were worse than those achieved when employing word count alone. However, this drop in performance was not observed for predictions of quality on the holdout-prompt set, where the models performed slightly better than the holdout-responses set.

Further, we observed that few-shot prompting of our Claude model could marginally surpass the performance of both GPT-2 and

Figure 4

Predicted Human Originality and Quality Scores for Few-Shot Llama-70b on the Holdout-Responses and Holdout-Prompt Sets



Note. Different colored points denote which item the examples were drawn from. Number of shots indicate the amount of examples provided to the model. See the online article for the color version of this figure.

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RoBERTa on the holdout-prompt set for originality ($r = .43$). Although few-shot prompting generally resulted in worse model performance than fine-tuning, findings for the holdout-prompt evidence a possible avenue for the automated originality scoring of CPST responses. This approach might be particularly useful for researchers who lack the copious amount of training data required to fine-tune LLMs. In training our fine-tuned models, we employed between 176 and 814 CPST responses for each item, rated with high agreement by multiple human raters. The need for large amounts of high-quality data is likely to make fine-tuning an inaccessible technique for many researchers, who might consider few-shot prompting as an alternative solution.

Correlations between model predictions of quality for both RoBERTa and GPT on the holdout-prompt set were the highest achieved in the entire analysis ($r = .89$). In other words, predictions for both models explained 79% of the variance in human ratings of quality. These results suggest that the models learned something general about how humans evaluate the quality of problem solutions that was consistent across problem prompts, but that originality evaluation may be more problem-specific. This observed difference may be due in part to the stronger relevance of superficial linguistic features such as word count in the prediction of quality. We nevertheless found that these models captured a degree of linguistic complexity beyond the simple elaboration of responses.

For both originality and quality, we found that model prediction errors—the extent to which model-predicted scores deviated from human scores—across both the holdout-responses and holdout-sets, were lower within two standard deviations than for responses at the extremes. This tendency to underestimate highly creative responses, and overestimate highly uncreative responses, is likely to stem in part from the LLMs inability to extrapolate real-world implications from text. Further, given that the data for both originality and quality scores tended to be normally distributed, the model was presented with fewer examples of responses associated with the extremes of this distribution. A similar observation was made in a recent attempt to automate the creativity scoring of drawings, indicating that this obstacle may be a pervasive one when it comes to the automated scoring of creativity (Patterson et al., 2024).

While LLMs are trained on vast quantities of natural language data and thus develop internal representations of meaning, they lack the extensive human experience that would allow for the consideration of highly contextualized responses. For originality, the high prediction errors for responses further from the mean are likely indicative of the way that human raters tend to score originality in accordance with response frequency. Given that the models are likely building an internal representation of the distribution of responses to the CPST during training, it is possible that they are engaging in a degree of frequency-based analysis when predicting originality scores. If this were to be the case, it then follows that they would underperform when presented with unseen items, as the models would lack an internal memory of the relevant responses and their frequency. Nevertheless, it is presently unclear why exactly do LLMs perform better when predicting human ratings of quality, rather than originality. Thus, further work should investigate this finding in depth, so as to maximize LLM performance and to further elucidate the different features of these two creativity metrics.

Of note, both RoBERTa and GPT-2 have been shown to underperform on several benchmark tasks when compared to more recent LLMs (e.g., L. Wang et al., 2023). Further, larger LLMs, with larger pretraining corpora tend to acquire emergent abilities that would previously appear unobtainable (e.g., reasoning; Hagendorff, 2023; Ichien et al., 2023; Wei et al., 2022). The current analysis should therefore be extended to larger models, such as GPT-4, to evaluate whether predictions of originality and quality on the CPST can be improved. It remains worth noting that at the time of writing this article some of these larger models like GPT-4 are not freely available for use, and researchers would likely incur considerable costs for employing these models in their work. In addition, we have not explored whether variations in the size of our training set would impact the performance of our fine-tuned models. As such, work is required to determine the minimum amount of CPST data required to maintain good model performance, or whether increasing the size of the training set might increase performance on the holdout-prompt set for originality. In a similar vein, then, it should be noted that our few-shot prompting approach is only an exploratory analysis which did not consider the vast array of hyperparameter settings and prompting strategies that can be employed for this purpose. Future work should thus extend the present analysis, by covering multiple combinations of hyperparameter and prompts to determine whether model performance can be increased in any way.

Further work should also investigate whether the present models present any bias toward groups that were underrepresented, or not included, in the CPST data sets. All CPST data sets were collected from undergraduate students enrolled at U.S. national universities, thus being likely composed of individuals from a skewed demographic background, such as all participants being highly educated. Indeed, the oversampling of individuals from such overrepresented demographic backgrounds is a widespread issue in psychological research (Henrich et al., 2010). Further, LLMs such as ChatGPT have been shown to align with American over other cultures (Atari et al., 2023). Specifically, ChatGPT was found to score more closely to the American population when tested on the World Values Survey, and more closely to people from Western, educated, industrialized, rich and democratic backgrounds when tested on a battery of cognitive tasks. As such, representative data sets should be collected and made openly available by researchers to ensure a comprehensive testing of model biases in future work.

Originality and quality are two distinct and important criteria for creativity (Reiter-Palmon, 2020), but automatic scoring methods have so far focused solely on originality. The present work demonstrates for the first time that automated creativity assessment can be successfully applied not only for originality, but also for quality. Further, we provide the first evidence that automated scoring can be applied to creative problem-solving tasks—tasks that emulate how creativity comes about in real-world settings such as the classroom or the workplace (Reiter-Palmon et al., 2009). Given the projected importance of creativity to the global economy and workforce (Florida, 2014), the development and dissemination of creativity tasks that can be efficiently scored is of vital importance.

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Appendix

Pearson's Correlations Between Human and Predicted Ratings of Originality and Quality, With and Without Outlier Removal

Human rating	Set	Predictor variable	Correlation with outlier removal (*r*)	Correlation without outlier removal (*r*)
Originality	Holdout-responses	RoBERTa	.79	.73
		GPT-2	.80	.74
	Holdout-prompt	RoBERTa	.41	.34
		GPT-2	.40	.37
Quality	Holdout-responses	RoBERTa	.83	.79
		GPT-2	.83	.80
	Holdout-prompt	RoBERTa	.89	.89
		GPT-2	.89	.89

Note. All Pearson's correlations significant at $p < .001$. RoBERTa = robustly optimized bidirectional encoder representations from transformers pretraining approach; GPT = generative pretrained transformer.

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